GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

MA.8.GR.1.2

I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

Warm-Up

1. How far apart are $(-2, -3)$ and $(4, 5)$?
2. Legs of 2 m and 7 m. The hypotenuse, to the nearest tenth?
3. A map's scale is 1 in : 30 mi. Towns 2.5 in apart are how far?
4. The rectangle $(-3, 2), (4, 2), (4, -1), (-3, -1)$: its perimeter?

Warm-Up

1. How far apart are $(-2, -3)$ and $(4, 5)$? **10 units**
2. Legs of 2 m and 7 m. The hypotenuse, to the nearest tenth?

   $\sqrt{53} \approx 7.3$ m
3. A map's scale is 1 in : 30 mi. Towns 2.5 in apart are how far?

   75 mi
4. The rectangle $(-3, 2), (4, 2), (4, -1), (-3, -1)$: its perimeter?

   20 units

Vocabulary

perimeter

the distance around a figure: the sum of all its sides

height of a triangle

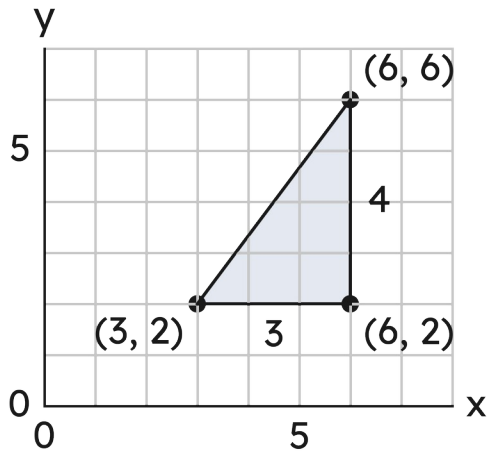
the segment from a corner straight to the opposite side, meeting it at a right angle

straight-line distance

the length of the one segment between two points — on a map, as the crow flies

Notes

I. Perimeter: every side — a slanted one by its triangle

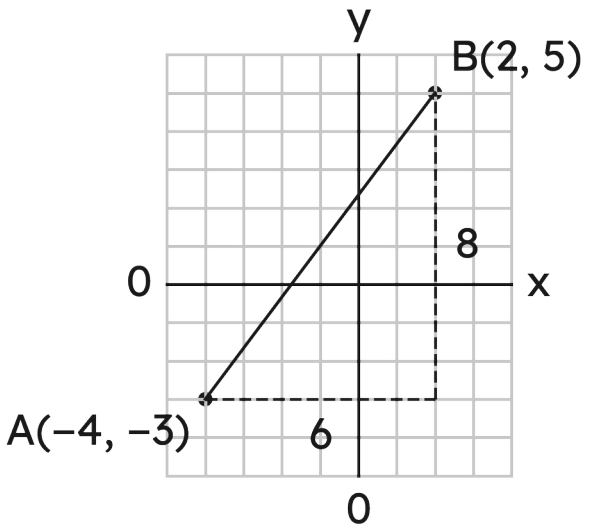


slanted side: $c^2 = 3^2 + 4^2 = 25, \quad c = 5$

$$P = 3 + 4 + 5 = 12$$

Notes

II. Along the streets, or straight

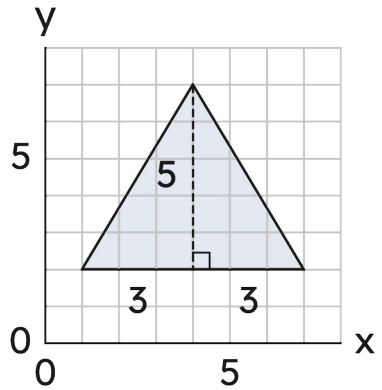


streets: $6 + 8 = 14$ blocks

straight: $c^2 = 36 + 64 = 100$, $c = 10$ blocks

Notes

III. No right angle: split it — and round once, at the end



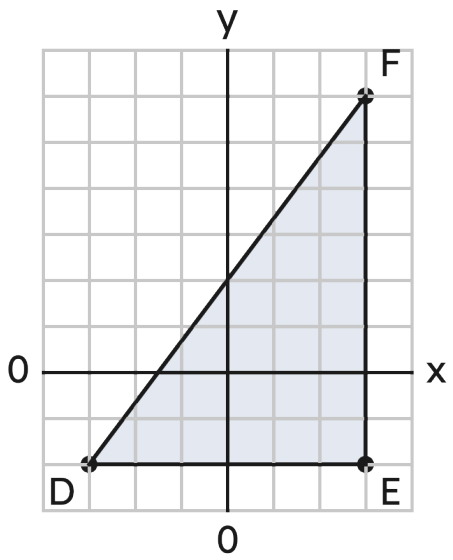
each slanted side: $c^2 = 3^2 + 5^2 = 34$, $c = \sqrt{34} \approx 5.831$

$$P \approx 6 + 5.831 + 5.831 = 17.662 \approx 17.7$$

Round once, at the end: 6 + 5.8 + 5.8 = 17.6 is short.

Example 1

Triangle DEF has corners $D(-3, -2)$, $E(3, -2)$ and $F(3, 6)$.



Find its perimeter.

Example 1

$3 - (-3) = 6, \quad 6 - (-2) = 8$

$c^2 = 36 + 64 = 100, \quad c = 10$

$P = 6 + 8 + 10 = 24$

Answer: 24 units

Your Turn

Find the perimeter of the triangle with corners

$(-5, -7)$, $(4, -7)$ and $(4, 5)$.

Your Turn

Find the perimeter of the triangle with corners

$(-5, -7)$, $(4, -7)$ and $(4, 5)$.

across $4 - (-5) = 9$, up $5 - (-7) = 12$

$c^2 = 81 + 144 = 225$, $c = 15$

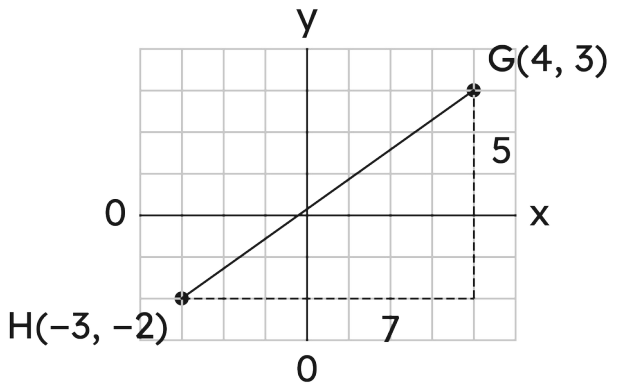
$P = 9 + 12 + 15 = 36$

Answer: 36 units

Example 2

On a town map in blocks, Ana's home H is at $(-3, -2)$

and the gym G at $(4, 3)$.



How far along the streets? In a straight line, to the nearest tenth?

Example 2

$4 - (-3) = 7, \quad 3 - (-2) = 5$

$7 + 5 = 12$

$c^2 = 49 + 25 = 74, \quad c = \sqrt{74} \approx 8.6$

Answer: 12 blocks along the streets; about 8.6 blocks straight

Your Turn

On a town map in blocks, Ben's home is at $(-5, 4)$ and the park at $(-2, -4)$. How far is it along the streets?

In a straight line, to the nearest tenth?

Your Turn

On a town map in blocks, Ben's home is at $(-5, 4)$

and the park at $(-2, -4)$. How far is it along the streets?

In a straight line, to the nearest tenth?

across 3, down 8: streets $3 + 8 = 11$

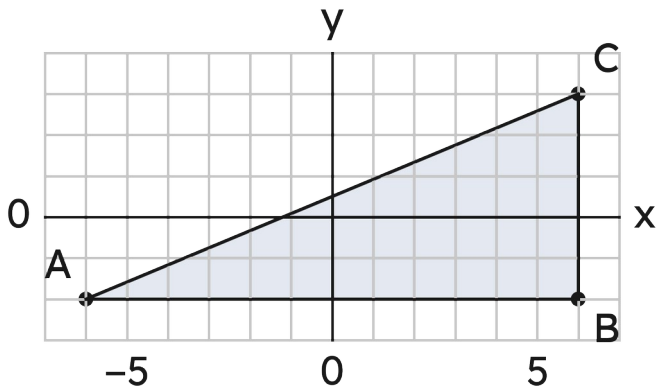
straight: $c^2 = 9 + 64 = 73$, $c = \sqrt{73} \approx 8.5$

Answer: 11 blocks along the streets; about 8.5 blocks straight

Whiteboards · Question 1 of 9

Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$.

Find its perimeter.

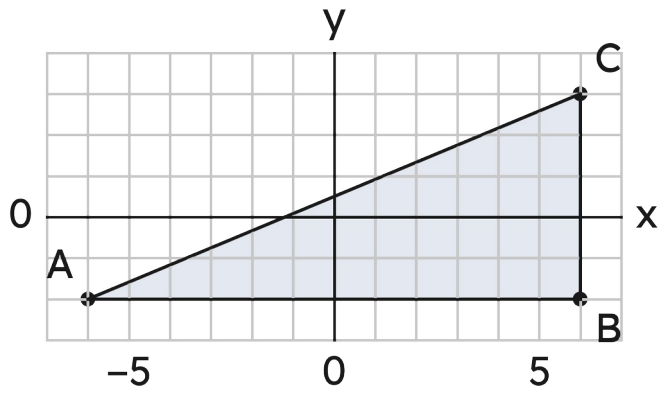


Answer it.

Whiteboards · Question 1 of 9

Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$.

Find its perimeter.



$6 - (-6) = 12$, $3 - (-2) = 5$

$c^2 = 144 + 25 = 169$, $c = 13$

$P = 12 + 5 + 13 = 30$

Answer: 30 units

Whiteboards · Question 2 of 9

A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$.

Find its perimeter, to the nearest tenth.

Answer it.

Whiteboards · Question 2 of 9

A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$.

Find its perimeter, to the nearest tenth.

$$1 - (-3) = 4, \quad 5 - (-4) = 9$$

$$c^2 = 16 + 81 = 97, \quad c = \sqrt{97}$$

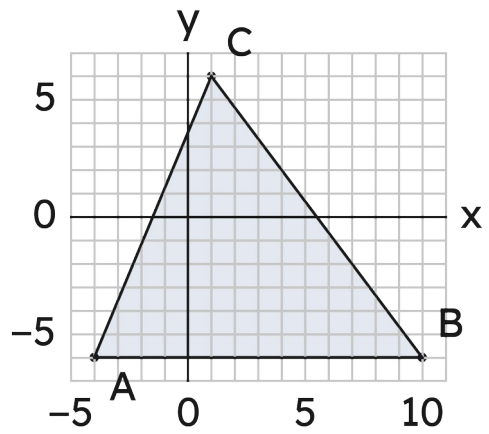
$$P = 4 + 9 + \sqrt{97} \approx 22.8$$

Answer: 22.8 units

Whiteboards · Question 3 of 9

Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle.

Find its perimeter.

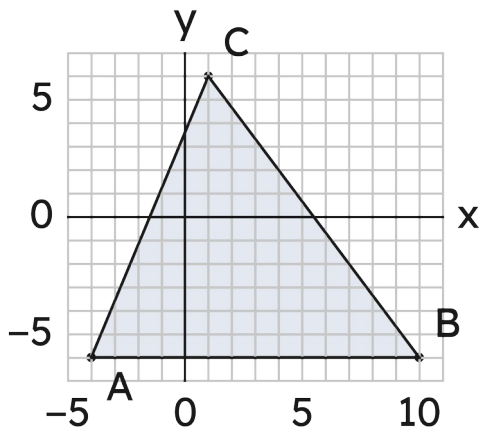


Answer it.

Whiteboards · Question 3 of 9

Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle.

Find its perimeter.



AC: $c^2 = 5^2 + 12^2 = 169$, $c = 13$

BC: $c^2 = 9^2 + 12^2 = 225$, $c = 15$

$P = 14 + 13 + 15 = 42$

Answer: 42 units

Whiteboards · Question 4 of 9

What is the perimeter of the triangle with corners $(-2, 3)$, $(2, 3)$ and $(2, -4)$, to the nearest tenth?

A. 11

B. 22

C. 76

D. 19.1

Answer it.

Whiteboards · Question 4 of 9

What is the perimeter of the triangle with corners $(-2, 3)$, $(2, 3)$ and $(2, -4)$, to the nearest tenth?

4 and 7 on grid lines

$c^2 = 16 + 49 = 65, \quad c = \sqrt{65}$

$P = 4 + 7 + \sqrt{65} \approx 19.1$

Answer: D — 19.1 units

Whiteboards · Question 5 of 9

On a map in blocks, the library is at $(-7, -1)$

and the pool at $(5, 4)$.

How much shorter is the straight line than the streets?

Answer it.

Whiteboards · Question 5 of 9

On a map in blocks, the library is at $(-7, -1)$

and the pool at $(5, 4)$.

How much shorter is the straight line than the streets?

streets: $12 + 5 = 17$

straight: $c^2 = 144 + 25 = 169$, $c = 13$

$17 - 13 = 4$

Answer: 4 blocks shorter

Whiteboards · Question 6 of 9

How long is a diagonal of the rectangle with corners $(-3, -4)$, $(6, -4)$, $(6, 8)$ and $(-3, 8)$?

A. 15

B. 21

C. 42

D. 10

Answer it.

Whiteboards · Question 6 of 9

How long is a diagonal of the rectangle with corners

$(-3, -4)$, $(6, -4)$, $(6, 8)$ **and** $(-3, 8)$**?**

sides $6 - (-3) = 9$ and $8 - (-4) = 12$

$c^2 = 81 + 144 = 225$, $c = 15$

Answer: A — 15 units

Whiteboards · Question 7 of 9

On a park map in meters, the fountain is at $(-5, 2)$

and the gate at $(4, -3)$.

How far is it straight across, to the nearest tenth?

Answer it.

Whiteboards · Question 7 of 9

On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$.

How far is it straight across, to the nearest tenth?

across $4 - (-5) = 9$, down $2 - (-3) = 5$

$c^2 = 81 + 25 = 106$, $c = \sqrt{106} \approx 10.3$

Answer: $\sqrt{106} \approx 10.3$ m

Whiteboards · Question 8 of 9

On a trail map in miles, the camp is at $(-3, -6)$,

the lake at $(2, 6)$ and the trailhead at $(-1, 0)$.

How far is the camp from the lake, in a straight line?

Answer it.

Whiteboards · Question 8 of 9

On a trail map in miles, the camp is at $(-3, -6)$,

the lake at $(2, 6)$ and the trailhead at $(-1, 0)$.

How far is the camp from the lake, in a straight line?

across $2 - (-3) = 5$, up $6 - (-6) = 12$

$c^2 = 25 + 144 = 169$, $c = 13$

Answer: 13 mi

Whiteboards · Question 9 of 9

Mia says the segment from $(0, 0)$ to $(3, 1)$ is 3 units long,

because it crosses 3 grid squares.

Is she right? Explain, and find its length to the nearest tenth.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Mia says the segment from $(0, 0)$ to $(3, 1)$ is 3 units long,

because it crosses 3 grid squares.

Is she right? Explain, and find its length to the nearest tenth.

across 3, up 1

$c^2 = 9 + 1 = 10, \quad c = \sqrt{10} \approx 3.2$

Answer: No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.

Independent Set

1. Find the perimeter of the triangle $(-2, -7)$, $(-2, 5)$, $(3, 5)$.
2. Find the perimeter of the triangle $(-4, 1)$, $(2, 1)$, $(2, -4)$, to the nearest tenth.
3. Find the perimeter of the parallelogram $(-4, -5)$, $(2, -5)$, $(7, 7)$, $(1, 7)$.
4. On a map in blocks, Kai's home is at $(-6, -5)$ and his school at $(3, 7)$. How much shorter is the straight line than the streets?
5. On a map in miles, a dock is at $(-4, 3)$ and an island at $(3, -5)$. How far apart are they, to the nearest tenth?
6. A rectangle has corners $(-3, -2)$, $(2, -2)$ and $(2, 10)$. Find its fourth corner and the length of a diagonal.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Pythagorean theorem: find the perimeter — VGE